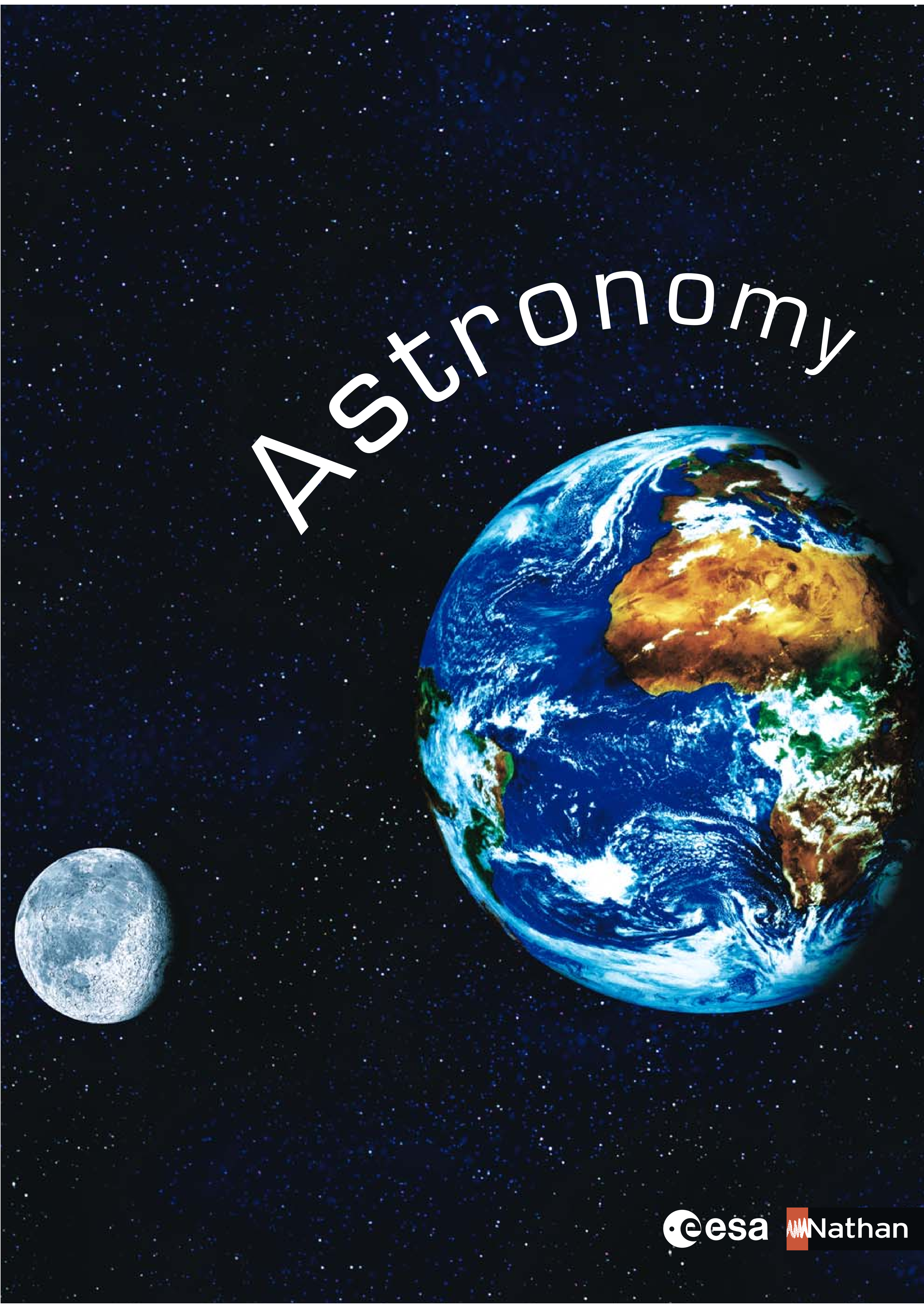
Astronomy



esa
Nathan

Astronomy

This document helps the teacher to support the study of everyday phenomena and encourage elementary school pupils to take the first steps towards constructing a scientific model.
It discusses the apparent movement of the Sun, the length of daylight and its changes in the course of the seasons, the rotation of the Earth on its axis, the Solar System and the Universe.

A BIT OF HISTORY

Astronomy is the science that studies the relative positions, movements, structure and evolution of celestial bodies. It grew out of everyday needs (measuring time, navigation, etc.) as well as human fears in the face of certain natural phenomena. For a long time, it was associated with astrology, the art of predicting events based on observation of the stars. Early astronomical knowledge was obtained by observing celestial phenomena visible to the naked eye for practical or religious purposes. In the 4th century BME (Before the Modern Era), the authority of **Aristotle**, a philosopher, imposed the belief that the Earth was immobile. Around 140 ME, **Ptolemy** described the Earth as a fixed point at the centre of the Universe. Classical astronomy came into being during the 16th and 17th centuries. In 1543, the astronomer **Nicolaus Copernicus** described the details of a system – which carries his name – in which the Earth, like the other planets, rotates on its own axis and around the Sun. In 1609, **Johannes Kepler** stated the first two laws of the movement of the planets in his work *Astronomia nova*. According to his studies, the planets move around the Sun in elliptical orbits; the closer they are situated relative to the Sun, the faster their orbit. In the same period, **Galileo** Galilei used a telescope that he made himself to observe the stars. He discovered the moons of Jupiter, the mountains and craters on the Moon's surface and the stars that form the Milky Way and he confirmed that the Earth is not at the centre of the Universe. After he published his findings, he was put on trial, and ordered by the Inquisitional Tribunal to recant his theories in 1633. In 1687, **Isaac Newton** established the fundamental laws of celestial mechanics by deducting from the work of Johannes Kepler and Galileo the principle of universal gravitation. This principle explains the movements of the stars: all bodies attract each other according to a force that depends on their mass and the distance between them. Thus, the greater the mass, the stronger its force of attraction. That is why the Earth holds the Moon in orbit and is itself held in orbit by the Sun; that is why an apple falls to the ground, attracted by the Earth. In 1670, Isaac Newton built a telescope using finely polished mirrors instead of traditional glass lenses, which enabled him to explore the area around the Sun. The general theory of relativity formulated by Albert Einstein in 1916 slightly corrected Newton's law. It explains the movements observed in the Universe by the fact that all bodies (stars, planets, galaxies, etc.) distort the space surrounding them.
Astronomy, which for a long time was limited to the Solar System, only gradually extended to the stars. In 1718, **Edmund Halley** discovered that the stars are not fixed to the vault of the heavens but have their own movements. In 1783, **William Herschel** recognised the movement of the Sun and in 1789, he built a spectacularly large telescope (12 metres long by 1.2 metres in diameter) which allowed him to observe double stars (one revolves around the other) and draw up a general map of the Milky Way.

Modern astronomy arose in the middle of the 19th century through the applications of photography and spectroscopy. In the first part of the 20th century, **Edwin Powell Hubble** used a telescope 2.5 metres in diameter to measure the distance of the galaxies relative to the Earth and their recession speed. He was one of the authors of the theory of an expanding Universe. In 1929, he formulated a law based on his observations: the greater the distance between any two galaxies, the greater the speed of their separation.
Starting from the theory of relativity formulated by Albert Einstein and the observations of Edwin Powell Hubble on the recession of the galaxies, in 1927, **Georges Lemaître** came up with the hypothesis of a gigantic primitive explosion, the Big Bang: all the galaxies in the Universe are moving away from each other, still carried by the enormous force of the blast. The expanding Universe theory is the currently accepted explanation for the origin of the Universe.

With the arrival of powerful telescopes, the widespread implementation of computerised calculation and the development of radio astronomy and space technology, astronomers continue to study the planets and their satellites, the Sun, the interplanetary environment, stars, nebulas, galaxies, etc.

Space technologies have contributed a new perspective to the study of the planets and the observation of celestial bodies. Space probes come within a short distance of the planets and their satellites and make measurements of them. So-called "astronomical" satellites carry instruments around the Earth used for observations that are transmitted by radio to astronomers.

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Space travel (aeronautics) has developed alongside advances in astronomy. In 1954, European countries joined forces in the space race by creating the European Space Research Organisation (ESRO) and the European Launcher Development Organisation (ELDO). The current European Space Agency (ESA) came into being in 1975.

ESA AND ITS SPACE ACTIVITIES

For nearly forty years now, the member countries of the European Space Agency have been pooling their resources to carry out a dynamic programme of space exploration and technological development. ESA sends missions to the far reaches of the Solar System and launches satellites to observe the Earth and improve telecommunications. It develops and possesses its own launchers, a spaceport and space laboratories.

The location of the ESA's spaceport in Kourou, French Guiana is ideal: its position close to the equator allows it to launch satellites into all types of orbits. The first Ariane 1 rocket was successfully launched there on 24 December, 1979. Since then, 166 launches have put satellites in orbit. Today, the Ariane 5 rocket can put several satellites in orbit at the same time. It will soon be joined by Vega, the small ESA launcher, and by the Russian launcher, Soyuz.

European space missions are not limited to sending satellites into orbit for purposes of Earth observation, meteorology, navigation and telecommunications. Some are sent into space to study our Solar System, the Universe, the stars, black holes, etc. The Hubble Space Telescope (HST) was jointly produced by the American space agency (NASA) and ESA. With a diameter of 2.4 metres, it has been sending back extraordinary images since 1990 and can perceive rays that are invisible to the naked eye.

Europe has carried out numerous scientific missions to learn more about the Universe, explain its origin and predict the evolution of the Solar System. The Huygens probe may find answers concerning the origin of our planet. In January 2005, it was parachuted through the atmosphere of Saturn to land on its largest moon, Titan, which has characteristics remarkably similar to those of the early Earth. As for the GAIA mission, it will map a billion stars to create a new three-dimensional atlas of our galaxy.

Another story is also taking place in space. In 1971, the first orbiting station, Saliout 1, was launched by the USSR, followed by the Russian space station MIR (1986-2001). Launched in 1998, the International Space Station (ISS) is a major scientific undertaking resulting from a joint endeavour of the United States, Russia, Canada, Japan and Europe. The ISS, which is as big as a football field, orbits around the Earth at an altitude of 400 km. Nine European astronauts have already visited it. These missions give us a better understanding of the laws of nature and certain physical phenomena. ESA has contributed to the ISS by providing the ATV, the Automated Transfer Vehicle, and Columbus, a space laboratory capable of hosting experiments in various fields (materials science, medicine, biology, technology).

GALILEO, the European satellite navigation system

THE EUROPEAN SPACE AGENCY (ESA) AND EDUCATION

The Education Department of the ESA develops educational material for the classroom aimed at increasing pupils' interest in science and technology. This material is intended for teachers in all disciplines.

It is available in several languages and may be freely used in the classroom.

Teaching avenues

ESA has two websites dedicated to the field of education:

The "EDUCATION" website: **www.esa.int/education** offers educational material (websites, kits, CDs, DVDs, etc.).

The "KIDS" website: **http://kids.esa.int**, in six languages, offers many short articles on space. Amusing activities and practical applications are available online and children will enjoy the numerous illustrations. Information recently taken from the main ESA website is adapted for young audiences and published every week under the "News" heading.

Some of the activities suggested in this supplement refer to these websites. They allow pupils to explore the websites on their own.

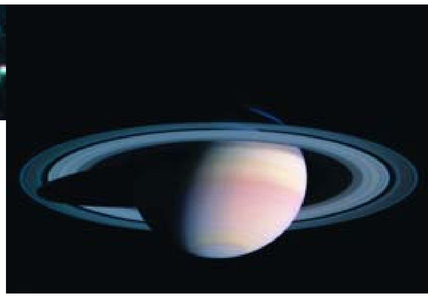
As part of pupils' training in the use of information and communication technologies, the teaching cards presented in the following pages enable teachers to consider the use of certain multimedia tools and the Internet and check their pupils' skills.

▸ The teacher can check to be sure that pupils know how to use the mouse and keyboard to select and enter data and navigate in a tree structure to search for specific information.

▸ Pupils must be able to consult a document on the screen – texts on the "Kids" website – and identify and sort the information contained in the document in order to engage in the proposed activities.

▸ The teacher may discuss the basic rules for the use of computers and the Internet and encourage pupils to take a critical appraoch towards the multitude of information offered on the ESA websites.

THE SOLAR SYSTEM AND THE UNIVERSE



© NASA

Saturn

When we observe the night sky, we can see a long whitish trail: **the Milky Way**. This is a galaxy – our galaxy – an agglomeration of nearly 200 billion stars, dust and gas, forming an enormous disk-shaped object. Inside the disk, near the edge, is the **Solar System**. Today we know there are millions of galaxies more or less like ours. They are often grouped together in clusters, which are themselves formed of clusters of clusters. These various groups make up the **Universe**.

The Solar System is a very small part of the Universe. It includes a central star, the **Sun**, with objects of different sizes, appearance and composition revolving around it: eight planets (according to the International Astronomical Union) and their satellites, comets, asteroids, etc. The orbits of these bodies are elliptical.

The objects forming the Solar System can be classified into six categories in terms of their chemical composition.

The **Sun**, which has a diameter of about 1.4 million km, is a ball of compressed gases (hydrogen and helium). It is the site of nuclear reactions that release energy, particularly in the form of light: it is this ability to shine by itself that makes the Sun a star.

There are four **terrestrial planets**: Mercury, Venus, the Earth and Mars, along with the Moon, the Earth's satellite. They are all solid and made of iron and silicates assembled into rocks. Their diameters vary from 300 to 12,000 km. They are called *terrestrial planets* because they have a solid, rocky surface similar to that of the Earth.

Asteroids are large rocks orbiting at a great distance from the Sun. Their composition is similar to that of the terrestrial planets, but they have smaller diameters of less than 100 km.

The **giant planets** can be observed beyond the asteroids. They are colder, less dense bodies of enormous size: Jupiter, Saturn, Uranus and Neptune. Like the Sun, they are balls of compressed gases (hydrogen and helium), but do not experience of nuclear reactions.

The **satellites of the giant planets**, as well as Pluto, are located close to these planets. With a smaller diameter, between 500 and 5,000 km, they are made of a mixture of ice and rock. There are two bodies at a great distance from the Sun – both of them satellites of Jupiter – that are exceptions: Io, which is similar to the rocky planets, and Europa, which is a rocky body covered with 100 km of water and ice.

The **comets** lying beyond Pluto, are clusters of ice containing rock and dust, with a diameter of less than 100 km. When a comet approaches the Sun, its surface heats up, releasing steam and dust that leave a characteristic trail tens of millions of kilometres long, known as the comet's tail.

In 2005, thanks to the observations of the Hubble telescope, a new planet was discovered in the solar system: Eris. It reflects more sunlight from its surface than almost any other body in the solar system.

According to the latest definition by the International Astronomical Union (IAU), approved on 24 August, 2006, "a **planet** is a celestial body that orbits around the Sun, with a mass of at least 5 x 10^{20} kg and a diameter of at least 800 km, which has cleared the neighbourhood around its orbit".

According to this definition, there are eight planets in our solar system: Mercury, Venus, the Earth, Mars, Jupiter, Saturn, Uranus and Neptune.

In addition, the IAU has created a new class of objects: dwarf planets including Pluto, Ceres and Eris. Pluto was previously considered the ninth planet. By extension, any celestial body that meets these criteria and gravitates around a star other than the Sun is called an *exo-planet*.

THE PLANET EARTH IN A FEW FIGURES

Distance between the Earth and the Sun	minimum – perihelion* in January maximum – aphelion* in July	147.1×10^6 km 152.1×10^6 km
Average orbital velocity around the Sun	between 29.29 and 30.29 km.s^{-1} (between 105,444 and 109,044 km.h^{-1})	
Length of a tropical year (between two solstices of the same nature)	365 days 5 hours 48 minutes 45 seconds	
Period of the Earth's rotation on its axis	23 hours 56 minutes 4 seconds	
Mass of the Earth	5.9736×10^{24} kg	
Volume of the Earth	1.083×10^{12} km^3	
Average temperature on the surface of the Earth	14 °C	
Tilt of the rotation axis	23°27′	
Composition of the atmosphere	78% nitrogen, 21% oxygen; 1% traces and other gases	

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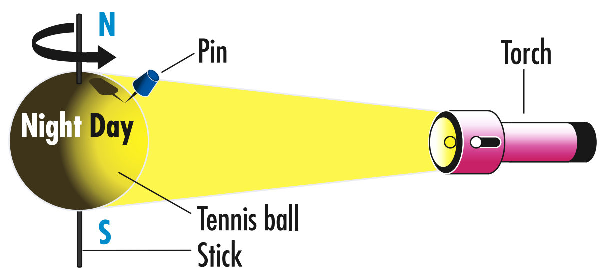
* *The perihelion is the point of a planet's orbit which is closest to the Sun.*
* *The aphelion is the point of a planet's orbit which is farthest from the Sun.*

For further classroom discussion

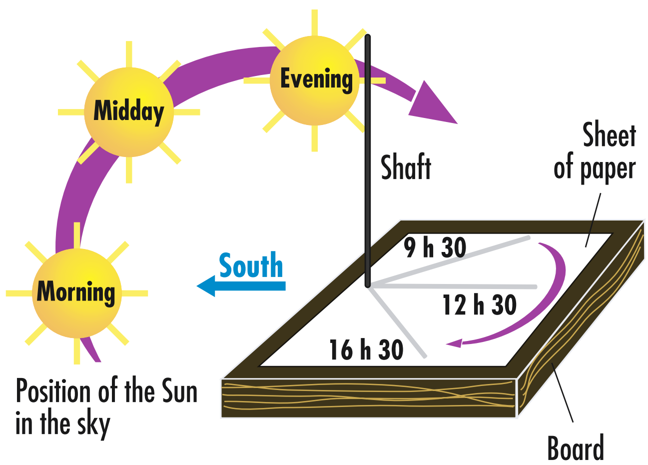
ALTERNATION OF DAY AND NIGHT

When the Sun moves in the sky, people everywhere in the world set their watches by its movements. But it is actually the Earth, which rotates its surface in front of the Sun. It takes 24 hours for the rays of the Sun to pass over the entire Earth. Therefore, one side of the Earth is lit by the Sun when the opposite side is in the dark: this is the alternation of day and night. To grasp this phenomenon, we can use a ball pierced with a stick. In the middle of the ball, a line drawn with a felt-tipped pen represents the equator. A torch simulates the Sun. Turn the ball in front of the torch. One side is lit up while the other is not.



THE APPARENT MOVEMENT OF THE SUN

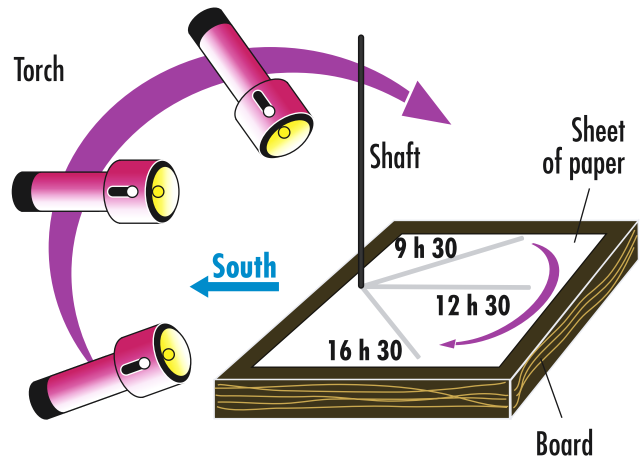
The **gnomon**, comprising a vertical shaft that casts a shadow on a flat surface, is the ancestor of the sundial. It is a useful tool for analysing the apparent movement of the Sun in a schoolyard.
The gnomon is used to visualise the shadow cast on a board by a thin shaft placed in the sun during the day. At different hours of the day, the shadow of the gnomon on the sheet of paper can be noted, to show the apparent movement of the Sun in the sky.



To do this reading properly, the following conditions must be satisfied:
– place the gnomon in a light, open place in order to follow the whole course of the Sun;
– orient the gnomon with the shaft facing South using a compass and make sure that its orientation does not change (e.g. mark its place with chalk);
– take a regular reading of the shadow on the sheet placed at the foot of the gnomon and note the time corresponding to the reading (one reading about every hour).

The pupils can observe that the shadow moves during the course of the day and that its size varies.

The movement of the shadows can then be reproduced in the classroom using a source of artificial light such as a **torch**. The pupils will therefore be able to observe in the darkness movement and the position of the light recreating the shadows. To reproduce the reading of the gnomon's shadows in the sun, the torch follows a curved trajectory. This trajectory corresponds to the apparent movement of the Sun.



ANSWERS FOR TEACHING CARDS

Teaching card 1

1. The Solar System is made up of the Sun and all the smaller objects that move around it.
2. **a)** Mercury, Venus, the Earth and Mars.
3. **a)** Jupiter, Saturn, Uranus and Neptune.
4. Some astronomers think it is too small.
5. Eris.
6. **a)** Jupiter.

Teaching card 2

1. Our planet is closest to the Moon. It is situated 380,000 kilometres from its satellite and 150 million kilometres from the Sun.
2. New moon: 1. First quarter: 2. Full moon: 3. Last quarter: 4.

Teaching card 3

2. **a)** This involves sending human beings into space for example to the Moon or towards Mars.
It is essential to ensure that the astronauts can travel there and back safely and survive on the planet.
b) Under the "Aurora" programme, ESA is working to find out the best way to send humans beyond Earth orbit. Exomars will put a wheeled robot on the planet which will explore the landing site.

CARD 1

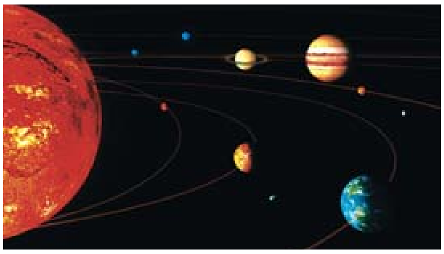
The Solar System

Surname ____________

First name ____________

Class ____________

Use the Internet address given by your teacher to find the ESA website. On the right-hand side, click on "Kids". On this page, select the flag of your country in the navigation bar, then click on "Our Universe", then, on the left-hand side, on "Planets and moons", finally, on the right-hand side of the screen, on "The Solar System and its planets".



The Solar System

1. What is the Solar System made up of?

....

....

2. **a)** Give the names of the four rocky planets closest to the Sun.

....

....

b) On the diagram of the Solar System below, fill in the name of these planets.

3. **a)** Give the names of the four gas giants of the Solar System.

....

....

b) On the diagram of the Solar System below, fill in the name of these planets.

4. Explain why Pluto is no longer considered a planet.

....

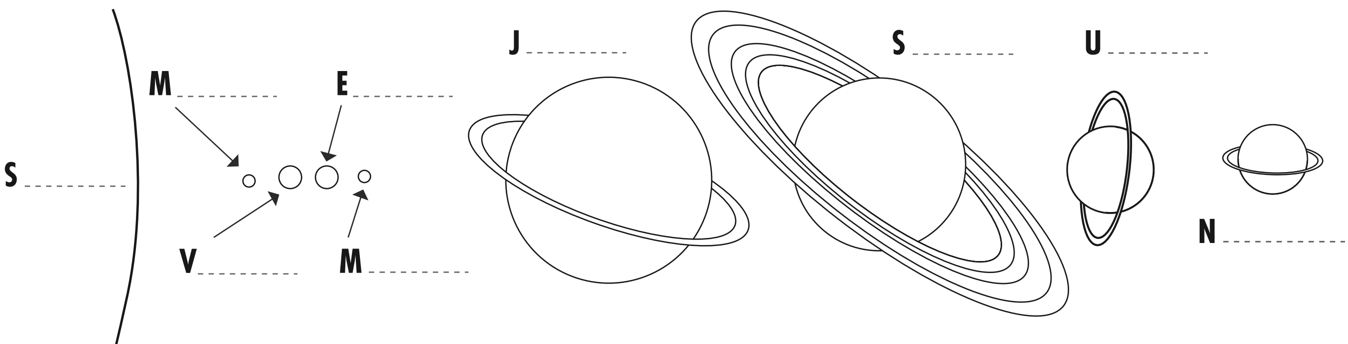
5. What is the name of the planet discovered in 2005 in the Solar System? ....

6. **Read the text below carefully, then follow the instructions.**

If the Sun were a big ball measuring 1.5 m in diameter, the planets would be the size of fruits. Jupiter would resemble a green melon (15 cm), Saturn a yellow grapefruit (13 cm), Uranus and Neptune two orange apricots (5.5 cm), the Earth and Venus two little red cherries (1.3 cm) and Mars and Mercury two white grapes (1.3 cm).

a) What is the largest planet in the Solar System? ....

b) On the diagram of the Solar System, colour each planet with the colour of the fruit.



CARD 2

The movements of the Earth and Moon

Surname ____________________

First name ____________________

Class ____________________

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The Moon and the Earth

1 The movements of the Earth

Read the text below carefully, then follow the instructions.

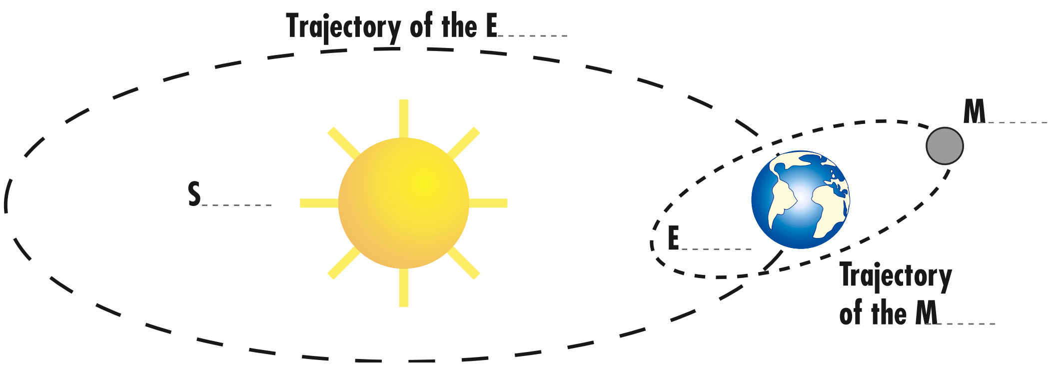
The Earth revolves around the Sun, which is situated 150 million kilometres from our planet. It makes this revolution in one year.
The Earth also rotates on its axis: it makes this revolution in 24 hours.
The Moon, situated at 380,000 kilometres from our planet, revolves around the Earth. It completes this revolution in about four weeks.

a) Is our planet closer to the Sun or to the Moon? Give a reason for your answer.

....

....

b) Using the information in the text, complete the diagram below using the following words: *Moon, Earth, Sun*. The same word may be used several times.

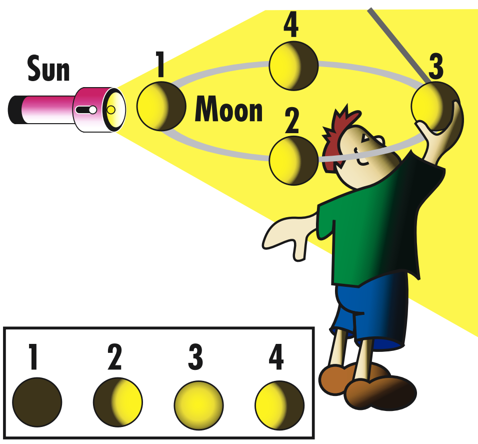


2 The phases of the Moon

Look carefully at the picture opposite.

Simulation of the movement of the Moon around the Earth

The Moon's appearance changes: these changes are called the phases of the Moon. On the diagram opposite, the torch represents the Sun, the ball represents the Moon and the pupil's head, the Earth. The pupil moves the ball around himself. Observe the shape of the lighted area of the ball and find the various appearances of the ball seen by the pupil below the diagram.



Match the various positions of the ball to the phases of the Moon.

Phase of the Moon in the course of a month	New moon	First quarter	Full moon	Last quarter
Position of the ball				

CARD 3

News from space

Surname ______

First name ______

Class ______

Use the Internet address given by your teacher to find the ESA website. On the right-hand side, click on "Kids".

Mars

1 On this page, select the flag of your country, then click on "News" on the left-hand side: this is recent information on the activities of the European Space Agency (ESA), which is updated every week.
Choose one news item from those displayed and read it carefully.

a) Give the title of the news item.

b) Indicate the date it appeared on the website.

c) Summarise the news item below in a few sentences.

2 In the top navigation area, click on "Life in Space", then on "Exploration".

a) Using the information on the page "Sending people to Mars…by the year 2035!", answer the following questions.

Explain what human spaceflight is.

What must be assured before sending humans to Mars?

b) Using the information on the page *ExoMars*, answer the following questions.

What is the purpose of the ESA programme called *Aurora*?

What is the aim of the ESA mission *ExoMars*?

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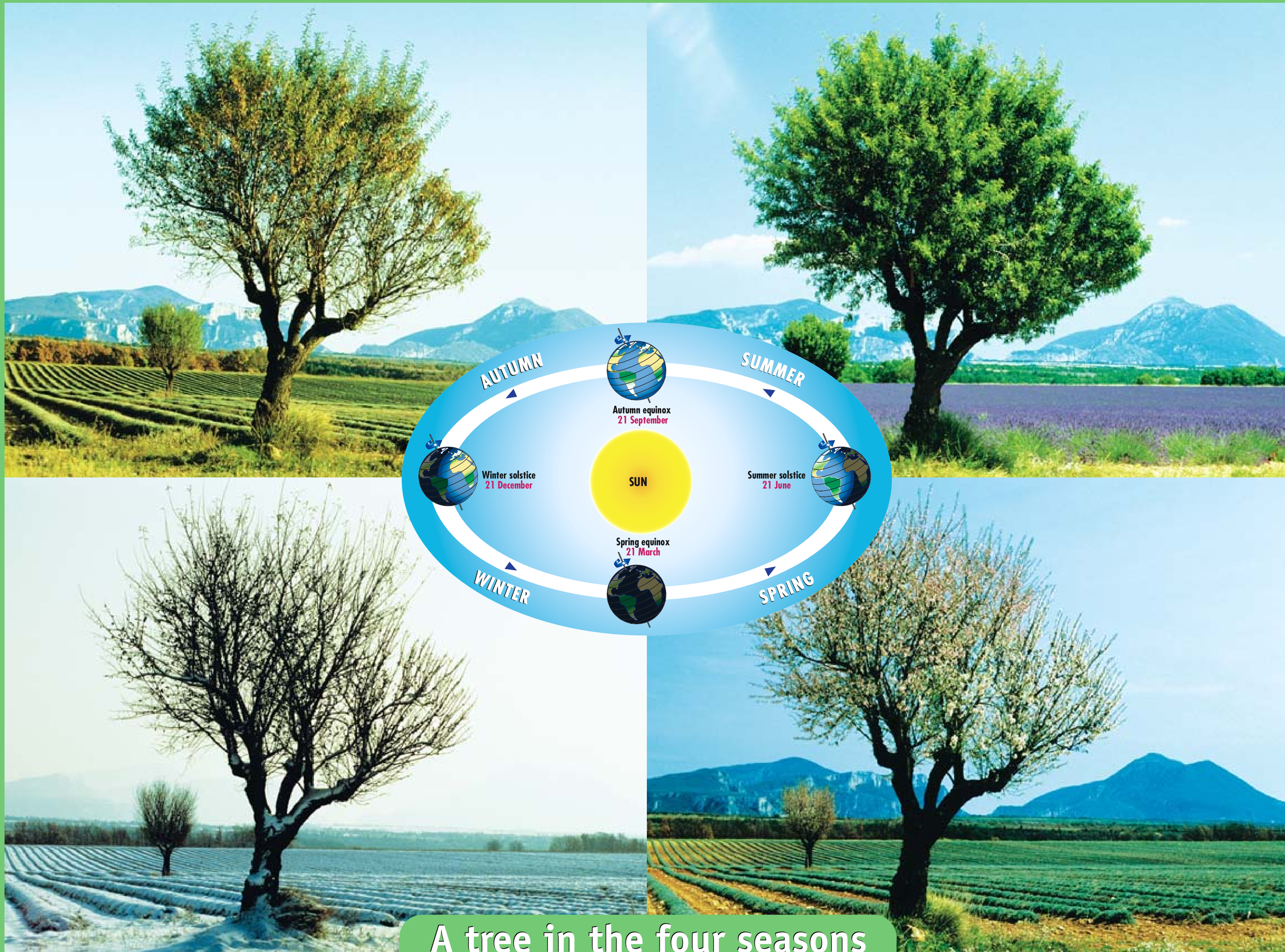
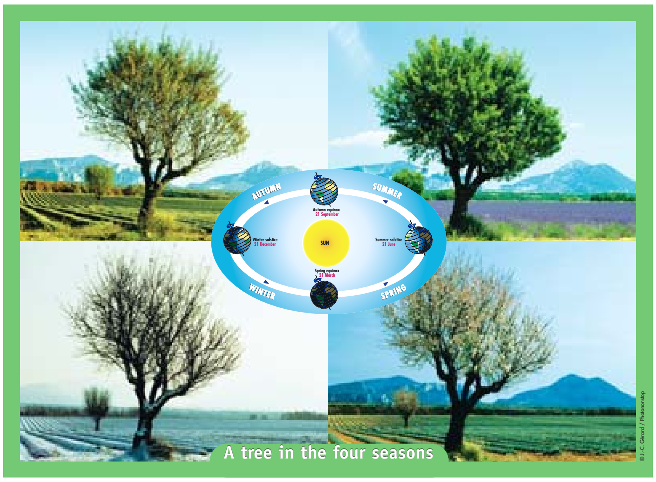


A tree in the four seasons

THE SEASONS

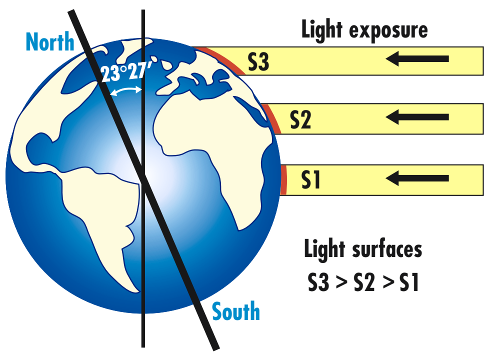


A tree in the four seasons

While the four photographs of the tree on the poster illustrate the four seasons, it should be remembered that the seasons are different depending on their location on the globe, due to the tilting of the Earth towards the Sun. In Teaching Card no. 1 of the poster, pupils can refresh their knowledge about the changing of the seasons on Earth. The phenomenon of the seasons results from the Earth's revolution around the Sun, its rotation on its axis and the tilt of its axis. The Sun orchestrates these various phenomena. Teaching Card no. 2 of the poster is devoted to this topic.

THE SUN, MASTER OF THE SEASONS

The changing of the seasons is due to the revolving movement of the Earth around the Sun and, more precisely, to the constant tilt of the Earth's axis of rotation relative to the perpendicular of its orbit plane as it revolves around the Sun (23°27′). In the course of its orbit, the Earth reaches the point closest to the Sun (the perihelion) in January and the point farthest from the Sun (the aphelion) in July. It is not the variation in the distance between the Earth and the Sun that causes the change of seasons, but rather the variation of the tilt of the Sun's rays relative to the Earth's surface and the number of hours of sunlight received by each hemisphere.
The angle made by the Earth's rotating axis and its normal orbital plane is fixed and equal to 23°27′. Consequently, as the Earth progresses in its orbit around the Sun, the orientation of the solar rays varies during the year, depending on latitude: the rays of the Sun touch the various points of the globe more or less vertically.
The greater the angle of exposure to the Sun's rays, the less energy is intercepted per unit area of the Earth's surface. The amount of energy received in a particular place will therefore vary during the seasons and, of course, during the day.
In the northern hemisphere, in summer, the Sun's rays are perpendicular to the ground and the days are long, because the arc described by the Sun above the horizon is long. In winter, the Sun remains low on the horizon, the rays are very oblique and the days are short. Summer is the warm season in the northern hemisphere and the cold season in the southern hemisphere.
The beginning of each season is defined by the **solstices** (summer and winter) and **equinoxes** (spring and autumn).
In the northern hemisphere, the period during which the rays are the most tilted is called the **winter solstice** on 21 or 22 December: it marks the beginning of that season. At the **summer solstice** on 21 or 22 June, the situation is reversed, and the North Pole is tilted towards the Sun. The greatest exposure to sunlight is in the Tropic of Cancer, where the Sun reaches its zenith at midday: this it is the beginning of summer.
At the equinox on 20 or 21 March and 21, 22 or 23 September, the days are of equal length. The Sun is at its zenith at midday at the equator. The energy is equally distributed over both hemispheres.
The dates of the solstices and equinoxes undergo fluctuations of up to three days depending on the year.



ANSWERS FOR TEACHING CARDS

Teaching card no. 1 of the poster

1. The first man lives near a pole (North or South), because there are only two seasons. The second man lives near the equator, because in his country, the days and nights are of equal length all year long. The third man lives in a temperate climate in the northern or southern hemisphere, with four, quite distinct seasons.

Teaching card no. 2 of the poster

1. **a)** The Sun gives light and heat to the Earth.
   b) Ultraviolet light causes sunburn and may cause cancer.
2. Diameter of the Sun: 1.4 million km, i.e. 109 times the diameter of the Earth.
   Mass of the Sun: 2 million, trillion, trillion, trillion kg. , i.e. 330,000 times the mass of the Earth.
   Distance between the Sun and the Earth: 150 million km.
3. **a)** It is made up of two gases: hydrogen and helium.
   b) Its temperature is 5,500°C.
   c) SOHO has been constantly watching the Sun for more than eight years.
   d) It sends images back to Earth which enable scientists to learn about the Sun and foresee “space weather” events that have consequences for our planet.
   e) An eclipse of the Sun only takes place at times of new Moon, when the Moon passes in front of the Sun and covers its brilliant face.

Poster CARD 1

Surname ______________________

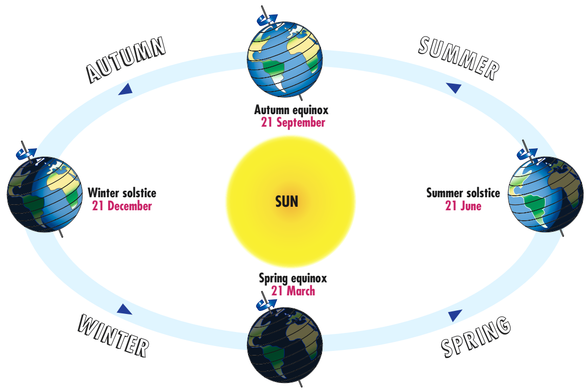
First name ______________________

Class ______________________

The Sun, master of the seasons

1 Look carefully at the document below, then read the text and answer the questions.

When one of the poles is tilted towards the Sun, it is day for six months. When it is tilted in the other direction, the night lasts six months.
At the equator, the Sun reaches its zenith every day of the year at midday. The days and nights are of equal length.
The more vertical the Sun's rays, the warmer the weather. The more the rays are tilted when they reach the ground, the colder the temperature.
The differences between the seasons are greater between the poles and the equator.
When the northern hemisphere is exposed to the most tilted rays, it is winter in that part of the world and it is summer in the southern hemisphere. And vice-versa.



An alien arrives on Earth and asks three men: "What is life like on your planet?"
"Simple", says the first man. "It is night for six months of the year and most of the animals hibernate. Then the sun rises and it become warm; the animals come out and plants start to grow. Six months later, the sun sets and night returns."
"Where I live", says the second man, "day and night last twelve hours each. It is the same all year long – Just a little thin rain at the end of the afternoon. It's monotonous!"
"That depends", says the third man. "In the summer, it is warm. In autumn, the trees lose their leaves. In winter, it is cold. And in the spring, the weather is fine. There are four different seasons."

Where do the three men encountered by the alien live? Justify your answer.

The first man: ....

....

The second man: ....

....

The third man: ....

....

2 Look up the definitions of the words "equinox" and "solstice" in the dictionary and copy them.

Equinox: ....

....

Solstice: ....

....

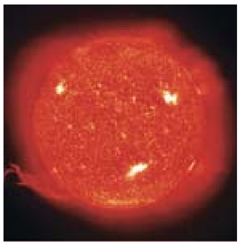
Poster CARD 2

Surname ____________

First name ____________

Class ____________

The Sun, a special star



The Sun

Use the Internet address given by your teacher to find the ESA website. On the right-hand side, move the cursor of the mouse over "Kids". On this page, select the flag of your country, then click on "Our Universe", then on the left-hand side, click on "The Sun" and finally, in the centre of the screen, click on "The Sun our star".

1 **a)** The Sun gives light and heat to the Earth.

....

....

b) What can the ultraviolet light produced by the Sun cause?

....

....

2 Complete the chart below:

Diameter of the Sun	.... km	i.e.	....	times the diameter of the Earth
Mass of the Sun	.... kg	i.e.	....	times the mass of the Earth
Distance between the Sun and the Earth	.... km			

3 **Next, click on each of the three links on the right hand side "The Sun", "Sun eclipses" and "SOHO". Read these pages carefully and answer the questions.**

a) Which elements make up the Sun?

....

....

b) What is its surface temperature? ....

c) SOHO has been constantly watching the Sun for more than eight years.

....

d) It sends images back to Earth which enable scientists to learn about the Sun…

....

....

e) An eclipse of the Sun only takes place at times of new Moon, when the Moon passes in front of the Sun and covers its brilliant face. ....

....

THE EUROPEAN SPACE AGENCY

ENVISAT, an Earth observation satellite

ESA's job is to draw up the European space programme and carry it through. The Agency's projects are designed to find out more about the Earth, its immediate space environment, the solar system and the Universe, as well as to develop satellite-based technologies and services, and to promote European industries. ESA also works closely with space organisations outside Europe.

In April 2007 ESA has 17 Member States. By coordinating the financial and intellectual resources of its members, it can undertake programmes and activities far beyond the scope of any single European country.

ESA's 17 Member States are Austria, Belgium, Denmark, Finland, France, Germany, Greece, Ireland, Italy, Luxembourg, the Netherlands, Norway, Portugal, Spain, Sweden, Switzerland and the United Kingdom. Canada, Hungary and the Czech Republic also participate in some projects under cooperation agreements.

ESA's headquarters are in Paris which is where policies and programmes are decided upon. ESA also has centres in a number of European countries, each of which has different responsibilities.

For example ESTEC, the European Space Research and Technology Centre located in Noordwijk, the Netherlands is the design hub for most ESA spacecraft and technology development.

In addition, ESA has liaison offices in Belgium, the United States and Russia; a launch base in French Guiana; and ground and tracking stations in various areas of the world.

The ESA website at **www.esa.int** has a wealth of information about ESA's daily work and programme results, available in all the languages of its Member States.

ESA SPACE NEWS

Europe on its way to Mars: the Aurora programme

The aim of the programme is to explore new destinations in space where traces of life might be discovered, first using robots, and later with manned space flights. Mars is one of those destinations.

By 2025, an international manned space flight to Mars will be possible. Returning to the Moon will be the preparatory phase for this flight. New technologies will have to be implemented to make this journey possible. Over the next twenty years, robotic missions will prepare the way for manned flights.

ESA Aurora programme plans to send robots as scouts. ExoMars, which is planned for launch in 2011, will be the first Aurora mission.

It will comprise a satellite and a landing module that can transport a large vehicle to the surface of Mars.

This vehicle, ExoMars Rover, will send several months crossing the rocky landscapes of Mars. The signals between it and the Earth will be relayed by the ExoMars satellite. The vehicle, powered by solar energy, will be able to drill into the ground, take samples of rocks and analyse them, search for water and study the local weather. Its primary aim will be to detect possible signs of life, past or present.

This programme will bring about important, innovative technological advances related to space probes, spaceship autonomy, re-entry into the Earth's atmosphere and the search for the presence of life. It will implement new technologies in the fields navigation and communication (precision landing and data transmission); propulsion; electric power production; data conversion, transmission, processing and storage. By its very nature, it will be an interdisciplinary venture, linking sciences and space technologies. Thus, the Aurora programme is designed as a roadmap for human exploration of space.

Virtual Observing of the Earth in real time: Miravi

The European Space Agency has created a website, MIRAVI, **www.esa.int/miravi** which provides access to the latest images sent by Envisat, the largest Earth observation satellite ever built. It contains daily-updated images of the Earth intended for scientists as well as the general public.

MIRAVI (which stands for Meris Image Rapid Visualization) follows the trajectory of Envisat around the Earth, generates images from the raw data collected by MERIS, the optical instrument of Envisat, and presents them online within two hours.

The Envisat mission is one of Europe's great successes. It is one of the main sources of information on the Earth and its environment, particularly regarding the factors affecting climate change. Since it was launched in 2002, Envisat has been continually observing land topography, the atmosphere, the oceans and the polar ice caps.

Envisat was put into polar orbit at an altitude of 800 km, which allows MERIS to observe the entire planet in three days.

In addition, images of the Earth, seen by various satellites including Envisat, are available at the following websites: **http://earth.esa.int/earthimages/** or **http://envisat.esa.int**.